Tasfia Anam



COMPUTER SCIENCE & ENGINEERING

House-33,Road-21,Block-B,Banani,Dhaka-1213, Bangladesh | tasfia.anam2002@gmail.com|
+8801632261227 | https://www.linkedin.com/in/tasfia-anam27/ | https://github.com/tasfiaanam27

PROFESSIONAL SUMMARY

Computer Science and Engineering student with hands-on academic project experience in machine learning, NLP, image processing, and web application development. Proficient in Python with foundational knowledge of AI/ML concepts, REST APIs, databases, and object-oriented programming. Familiar with deep learning and RAG-based systems through academic projects. Strong analytical and problem-solving abilities with a passion for learning modern AI technologies and software development practices.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java (Basic)
Web Technologies: HTML, CSS, JavaScript
Database: SQL
Tools & Platforms: GitHub, Visual Studio Code, Google Colab, Anaconda
Core Knowledge: Data Structures, Algorithms, DBMS, Object oriented Programming, Computer Networking, Software Development Life Cycle, Image Processing
Frameworks and Libraries: OpenCV, NumPy, Pandas, Matplotlib, Django

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science and Engineering
Independent University, Bangladesh | 2021- Expected 2026
CGPA: 3.34
Completed all academic coursework; thesis defense pending.

Advanced Level (A Level)
Lords-An English Medium School | 2019-2021

Ordinary Level (O Level)
Academia | 2018-2019

PROJECTS

RAG-Based NLP User Story Refinement Framework

Technologies: Python, NLP, Machine Learning, Retrieval-Augmented Generation (RAG), NLTK, Pandas, Vector Database

- Developed an NLP-based framework to transform unstructured Agile user stories into structured, standardized, and testable formats.
- Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline combining text preprocessing, vector-based retrieval, and LLM-assisted refinement.
- Created and evaluated a synthetic dataset (5K+ samples, 1K annotated pairs) to improve model robustness and consistency.
- Evaluated output quality using BLEU and ROUGE metrics, improving clarity and reducing ambiguity in generated user stories.

Bone Fracture Classification Using Deep Learning and Image Processing Techniques

Technologies: Python, TensorFlow, OpenCV, NumPy, Matplotlib, ResNet50

- Applied deep learning and computer vision techniques for medical image classification and performance analysis.
- Implemented image preprocessing techniques including normalization, filtering, and augmentation to improve model performance.
- Trained a CNN model using transfer learning with ResNet50 for fracture classification.
- Evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrix analysis.

SmartSadaqah – AI-Based Healthcare Donation System

Technologies: Python (Django), HTML, CSS, JavaScript, SQLite, REST APIs

- Developed a full-stack web application to connect donors with verified healthcare funding projects.
- Worked on backend logic, database integration, authentication, and system workflow implementation using Django.
- Built a responsive frontend interface using HTML, CSS, and JavaScript for user interaction and donation tracking.
- Integrated a rule-based AI module for donation allocation and project recommendation to improve transparency and efficiency.

EXPERIENCE

Counsellor, Application Team
PIE International Education | April 2022 – September 2022

- Coordinated with students and international representatives to support application processing and documentation workflows.
- Managed records, resolved operational issues, and maintained effective communication within cross-functional teams.
- Developed organizational, problem-solving, and stakeholder communication skills in a fast-paced environment.

CERTIFICATIONS

- CCNAv7: Introduction to Networks – Cisco Networking Academy (Cisco), 2024
- Duke of Edinburgh Award – Bronze, Silver, Gold Levels – The Duke of Edinburgh's International Award
- Live-in Field Experience (LFE) Certification – IUB Field Learning Program

LANGUAGES

- English (Professional Proficiency)
- Bangla (Native Proficiency)